

WIRSCHING’S WEAK COVERING CONJECTURE: AN EXTENDED COMPUTATION, A PROVEN BOUND, AND THE STRUCTURE OF WHAT RESISTS IT

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ABSTRACT. Wirsching’s 1998 Weak Covering Conjecture asks for a sub-exponential $K(l)$ such that a large enough set $R_{j-1,j} \subset \mathbb{Z}_3^*$, built from strictly decreasing exponent tuples, is forced to cover every unit residue modulo 3^l . We extend the exact computation of $j^*(l)$, the smallest such covering budget, from $l = 20$ to $l = 23$, and compare four growth models for $e(l) := j^*(l) - l \log_4 3$ against the extended table: a constant model is disfavored at every statistic checked, but logarithmic, square-root, and slow-linear growth remain statistically indistinguishable from one another. We prove the best unconditional bound known to us, $j^*(l) \leq \frac{9}{8}l + O(1)$, via a mean-payoff-game relaxation of the covering budget. We then examine the natural Fourier approach to a tighter bound: a uniform square-root-cancellation argument is capped below even this bound, a direct computation at one accessible level finds the bulk of the relevant exponential sum’s mass sitting outside any sparse exceptional set, and the residues that actually fail to be covered sit at extremal cells that a phase-randomization experiment shows are decided by phase structure that magnitude alone does not capture. We prove a theorem on the uncovered residues themselves: the set left uncovered at one budget is contained in the union of its double and its quadruple at the next, giving an explicit recursive upper bound on $j^*(l)$ that is empirically exact at every budget checked. We falsify the natural cost-1 form of a local repair rule this theorem might suggest, with explicit witness-level exhaustive counts, prove that every last-uncovered residue is $\equiv 1 \pmod{3}$, and report several further exact identities, verified but not proven, governing the last residues to be covered. Two precisely stated open questions remain, each narrower than the conjecture itself. The Weak Covering Conjecture remains open.

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1. INTRODUCTION

Wirsching’s 1998 monograph [1] studies the density of predecessor sets under the $3n + 1$ map through an averaging construction on the 3-adic integers. A limiting Markov chain built from that construction has an invariant density, and Wirsching’s own route to positive predecessor density passes through a covering question about a family of finite subsets of $\mathbb{Z}_3^*$, the 3-adic units. For integers $j, k \geq 0$, write

$$(1) \quad R_{j,k} := \left\{ \sum_{i=0}^j 2^{a_i} 3^i \mid j+k \geq a_0 > a_1 > \cdots > a_j \geq 0 \right\} \subset \mathbb{Z}_3^*.$$

Each element of $R_{j,k}$ is built from a strictly decreasing sequence of $j + 1$ “digit positions” chosen from $\{0, \dots, j + k\}$: the i -th chosen position, counting from the top, contributes 2^{a_i} scaled by 3^i . Wirsching’s Weak Covering Conjecture (Conjecture 3.9 of [1]) asks for a sub-exponential $K(l)$ such that $|R_{j-1,j}| \geq K(l) \cdot 3^l$ already forces $R_{j-1,j}$ to cover every unit residue modulo 3^l ; the cardinality on the left is that of the tuple family itself, before reduction, which by Proposition 1’s bijection equals $\binom{2j}{j}$, not the (a priori much smaller) size of its image modulo 3^l . Equivalently, writing $j^*(l)$

for the smallest j such that $R_{j-1,j}$ covers $(\mathbb{Z}/3^l\mathbb{Z})^*$ exactly, the conjecture concerns the growth rate of

$$(2) \quad e(l) := j^*(l) - l \log_4 3,$$

the excess of the true covering budget over the leading-order rate a cardinality argument alone predicts ($|R_{j-1,j}| = \binom{2j}{j} \sim 4^j$, so matching $2 \cdot 3^{l-1}$ residues needs $j \sim l \log_4 3$ at leading order).

An earlier, unpublished manuscript of the author computed $j^*(l)$ exactly for $l = 1, \dots, 20$ by direct enumeration and reported the resulting table without a growth-rate analysis. This paper does three things. First, it extends that table to $l = 23$ and gives a proper statistical comparison of candidate growth models for $e(l)$ against the full range (Sections 2 and 3). Second, it gives the best unconditional upper bound on $j^*(l)$ we have found, via a reformulation of the covering budget as a mean-payoff game (Section 4). Third, and at greatest length, it reports a sustained attempt to close the gap to the conjectured rate directly, first by the natural Fourier approach, and, once that approach is shown to fail for a precise and checkable reason, by a direct study of the residues that actually resist coverage (Sections 5 and 6). That second attempt does not settle the conjecture. It produces a theorem on those residues, a falsification of the cost-1 form of a repair rule the theorem might suggest, and two precisely named open questions, each narrower than the conjecture itself and worth a dedicated attack in its own right.

2. THE COMPUTATION

Table 1 gives $j^*(l)$ and $e(l)$ for $l = 1, \dots, 23$. The values for $l \leq 20$ reproduce the earlier manuscript's table exactly; they were independently recomputed here by a from-scratch Rust reimplement of the covering search (a bitset rotation dynamic program over residues modulo 3^l), validated against the original Python implementation at every $l \leq 20$ before being trusted for the extension. The values for $l = 21, 22, 23$ are new.

l	$j^*(l)$	$e(l)$	l	$j^*(l)$	$e(l)$	l	$j^*(l)$	$e(l)$
1	1	0.208	9	13	5.868	17	21	7.528
2	4	2.415	10	15	7.075	18	22	7.735
3	6	3.623	11	16	7.283	19	23	7.943
4	7	3.830	12	17	7.490	20	24	8.150
5	9	5.038	13	18	7.698	21	25	8.358
6	10	5.245	14	19	7.905	22	26	8.565
7	11	5.453	15	20	8.113	23	27	8.773
8	12	5.660	16	20	7.320			

TABLE 1. $j^*(l)$ and $e(l) = j^*(l) - l \log_4 3$, $l = 1, \dots, 23$.

The single plateau in the table, $j^*(15) = j^*(16) = 20$, is not an error: $l = 15 \rightarrow 16$ is the only increment among the 22 steps from $l = 1$ to $l = 23$ at which the covering budget does not increase, and it is exactly the event Section 3's plateau-frequency test measures.

$l = 21$ fits in physical RAM once a memory-wasteful parallelization step in the original implementation was corrected. $l = 22$ required 500 GiB of swap, added to the machine specifically for this computation, and took approximately 10,749 s (~ 3 hours) with swap I/O as the dominant cost, against 748 s for $l = 21$ without swap. $l = 23$ required approximately 263 GiB of state and 375,615 s (~ 104 hours); the successful run had no interruption, so a checkpoint/resume mechanism added after an earlier, unrelated policy-triggered kill of an interrupted attempt at the same level was never exercised on it. Three attempts at $l = 24$, made over the course of this project on the same machine, each failed before completion (two to memory-policy kills, one to an apparent out-of-memory condition with no surviving kernel log confirming it directly); $l = 24$ is not attempted further, and the table stops at $l = 23$.

3. THE GROWTH OF $e(l)$

We compare four candidate functional forms for $e(l)$'s growth against the tail $l = 10, \dots, 23$ ($n = 14$; the same range choice as the earlier manuscript, now with three more points): a constant (stabilization), a logarithmic term, a square-root term, and a slow-linear term. Table 2 gives the fit statistics.

model	k	ΔAIC	ΔBIC	LOOCV RMSE
constant	1	16.091	15.452	0.5200
logarithmic	2	1.512	1.512	0.2991
square-root	2	0.722	0.722	0.2909
slow-linear	2	0.000	0.000	0.2837

TABLE 2. Model comparison on $e(l)$, $l = 10, \dots, 23$. $\Delta\text{AIC}/\text{BIC}$ relative to the best-fitting model (slow-linear).

A constant model is disfavored sharply and consistently: $\Delta\text{AIC} = 16.09$, $\Delta\text{BIC} = 15.45$, both growing at every l added since $l = 21$. The three growth models remain statistically indistinguishable from one another by the conventional $\Delta\text{AIC} > 2$ rule of thumb, with slow-linear taking a narrow lead on both AIC and leave-one-out cross-validation; the three candidate regressors ($\log l$, $\sqrt{l}$, l) are themselves highly correlated over this range (pairwise correlation ≥ 0.993), so this ranking says more about how thin the margin currently is than about which functional form is correct.

A direct combinatorial test tells the same story from a different angle. Among the 22 increments $j^*(l+1) - j^*(l)$ for $l = 1, \dots, 22$, exactly one is zero (the $l = 15 \rightarrow 16$ plateau noted above). The earlier manuscript's own table, $l \leq 20$, showed 3.9 plateaus expected in 19 increments under a constant- $e(l)$ rounding model; taking $q = 3.9/19 \approx 0.2053$ as that model's plateau probability and testing the observed count against $\text{Bin}(22, q)$, one-sided, gives $p = 0.0426$, crossing the conventional 0.05 threshold for the first time with the $l = 23$ point included; restricted to the same tail the model comparison above uses ($n = 13$), the same test gives $p \approx 0.22$. We report both figures: $e(l)$ is an exact, deterministic sequence, so neither AIC/BIC nor this test carries the sampling interpretation of a classical inference over noisy data. Both are best read descriptively, as "how well does a constant- $e(l)$ source fit the observed increment pattern."

A constant $e(l)$ is disfavored statistically by every measure above. It is also, unconditionally, impossible.

Proposition 1 (Unboundedness of $e(l)$). $e(l) \geq \frac{1}{4} \log_2 l - O(1)$.

Proof. Distinct exponent tuples give distinct sums: the smallest exponent a_{j-1} is recovered from $V = \sum_i 2^{a_i} 3^i$ as its 2-adic valuation, $2^{a_{j-1}}$ is then subtracted off, and the process iterates on the remaining $j-1$ terms. So $R_{j-1,j}$, in bijection with j -subsets of the $2j$ -element set $\{0, \dots, 2j-1\}$, has exact cardinality $\binom{2j}{j}$. Covering $(\mathbb{Z}/3^l\mathbb{Z})^*$, of size $2 \cdot 3^{l-1}$, requires the domain to be at least as large as the codomain: $\binom{2j}{j} \geq 2 \cdot 3^{l-1}$ is necessary for $j \geq j^*(l)$. The explicit bound $\binom{2n}{n} \leq 4^n / \sqrt{3n+1}$, valid for every $n \geq 1$, turns this into $4^j \geq 2 \cdot 3^{l-1} \sqrt{3j+1}$, hence

$$j \geq l \log_4 3 - \log_4 3 + \log_4 2 + \frac{1}{2} \log_4 (3j+1).$$

Corollary 3 below already gives $j = O(l)$, so $\log_4 (3j+1) = \frac{1}{2} \log_2 l + O(1)$, and the claimed bound follows. $\square$

4. A MEAN-PAYOFF-GAME UPPER BOUND

Reading an admissible tuple's exponents from the bottom up encodes covering as a game of imperfect information. A state is a unit $z \in \mathbb{Z}_3^*$. A move of cost $d \geq 0$ is *legal* at z when $2^d z \equiv 1$

(mod 3) (so that $T_d(z) := (2^{d+1}z - 2)/3$ is an integer), and *safe* when every one of the three residues $z, z+m, z+2m$ modulo $3m$, for the relevant modulus m , maps under T_d back into a unit. A minimizing player, seeing only $z \bmod 3^k$, chooses a legal, safe d ; an adversary then reveals the hidden next 3-adic digit $\epsilon \in \{0, 1, 2\}$, forcing the true state to $T_d(z+3^k\epsilon) \bmod 3^k$. Both requirements matter: legality alone can send a unit to 0 (take $z = 1, d = 0$: $T_0(1) = 0$), and the safety filter is exactly what keeps the walk inside $\mathbb{Z}_3^*$ regardless of which digit the adversary supplies. That every unit state admits at least one legal, safe move has been verified computationally at every $k = 3, \dots, 14$ used below, not proven in general here.

This restricted-visibility game is, by construction, a finite, two-player *mean-payoff game* on the $2 \cdot 3^{k-1}$ unit states modulo 3^k : legal, safe moves are the minimizer's edges, the adversary's three choices of ϵ are the maximizer's edges, and the payoff is the long-run average of the move costs d . The correspondence between this game (at full precision, $k = l$) and the original covering problem, and hence the equality of the two objects' finite-horizon values, was verified computationally against direct brute-force enumeration of $R_{j-1,j}$'s covering sets for $l = 1, \dots, 6$, with no discrepancy; the two constructions are not independently conjectured to agree, they are checked to.

Theorem 2 (Window relaxation bound). *For every $k \geq 1$ there is an explicit rational constant C_k such that $j^*(l) \leq \rho_k \cdot l + C_k$ for all l , where ρ_k is the value of the window- k game above, and ρ_k is non-increasing in k (a window- k policy is a special case of a window- $(k+1)$ policy).*

Proof. By the classical theorem of Ehrenfeucht and Mycielski [2], every mean-payoff game on a finite graph has a value ρ_k , together with an optimal positional minimizing policy carrying a potential (bias) function h satisfying, at every state s and for every adversary digit ϵ ,

$$d(s) + h(s'(\epsilon)) \leq \rho_k + h(s),$$

where $s'(\epsilon)$ is the successor under the minimizer's chosen move. Summing this inequality along any l -step play $s_0, s_1, \dots, s_l$ telescopes to

$$\sum_{i=0}^{l-1} d(s_i) \leq l\rho_k + h(s_0) - h(s_l) \leq l\rho_k + (\max h - \min h) =: l\rho_k + C_k,$$

regardless of which digits the adversary actually supplies, since the potential inequality holds for every ϵ at every step. The window- k policy needs only $z \bmod 3^k$ to choose each move, so it is executable against the true, full-precision covering problem; by the verified correspondence above, $\sum d(s_i)$ over l steps started at $z \bmod 3^k$ upper-bounds the covering budget needed for z modulo 3^l . Taking the worst-case z gives $j^*(l) \leq \rho_k l + C_k$. $\square$

ρ_k and C_k are computed exactly, as rationals, by a nested Howard strategy-improvement solver, self-certified at every k via a matching adversary lower bound extracted from the same run (upper bound equals lower bound, so the computed rational equals ρ_k exactly), and cross-validated by an independent value-iteration solver and, at $k = 4$ and $k = 7$, by running the resulting policy against the actual finite covering DP for every unit residue modulo 3^l , l up to 14, with zero illegal or unsafe moves encountered. Table 3 gives the values computed for $k = 3, \dots, 14$.

k	3	4	5	6	7	8	9	10	11	12	13	14
ρ_k	2	$\frac{5}{3}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{7}{5}$	$\frac{25}{19}$	$\frac{5}{4}$	$\frac{11}{9}$	$\frac{6}{5}$	$\frac{7}{6}$	$\frac{119}{104}$	$\frac{9}{8}$

TABLE 3. The window-relaxation value ρ_k , $k = 3, \dots, 14$.

At $k = 14$, $\rho_{14} = 9/8 = 1.125$, giving:

Corollary 3. $j^*(l) \leq \frac{9}{8}l + O(1)$.

This is, to our knowledge, the best unconditional bound on $j^*(l)$'s growth rate currently established. The general technique, representing a covering-type combinatorial problem as a mean-payoff game and computing a decreasing sequence of exact rational relaxation bounds, is not new to this application: beyond Ehrenfeucht and Mycielski's foundational theorem [2] itself, a close and recent methodological precedent proves rationality and computability of the covering radius of a primitive sofic shift by essentially the same construction, applied to a different object [3]. We cite both as precedent for the framing; what is new here is its specific application to $j^*(l)$.

Remark 4. Fitting $\rho_k = L + A/k$ by ordinary least squares to $k = 10, \dots, 14$ gives $L \approx 0.876$, still well above the conjectured true rate $\log_4 3 = 0.7925\dots$; the fit is sensitive to the range used (all of $k = 3, \dots, 14$ gives $L \approx 0.904$) and is not evidence of any specific limit. Only the direction $j^*(l) \leq \rho_k l + C_k$, proven above, is established. The reverse inequality, $\inf_k \rho_k \leq \limsup_l j^*(l)/l$, which would be needed to conclude anything about where ρ_k is heading, is not established, and there is evidence in the course of this work that the window relaxation may carry an irreducible gap above the true rate.

5. THE EXPONENTIAL SUM, AND WHY IT RESISTS A DIRECT ATTACK

Corollary 3's bound, $1.125l$, is well above the conjectured rate $\log_4(3)l \approx 0.7925l$. The natural route to closing that gap is Fourier-analytic. By orthogonality on $\mathbb{Z}/3^l\mathbb{Z}$, the number of tuples landing in residue class z is

$$(3) \quad N(z) = \frac{1}{3^l} \sum_{t=0}^{3^l-1} S(t) e\left(-\frac{tz}{3^l}\right), \quad S(t) := \sum_{j+k \geq a_0 > \dots > a_j \geq 0} e\left(\frac{t}{3^l} \sum_{i=0}^j 2^{a_i} 3^i\right),$$

and, since $S(0) = |R_{j,k}| =: T$, the triangle inequality gives $N(z) > 0$ for every z as soon as $\sum_{t \neq 0} |S(t)| < T$; a uniform bound $|S(t)| = o(T/3^l)$ at every nonzero t would suffice, at a rate controlled directly by the exponent in the bound. This approach does not work, for three separate and mutually reinforcing reasons.

5.1. A hard ceiling on uniform cancellation.

Proposition 5. *Any proof of covering via a uniform bound $|S(t)| \lesssim \sqrt{T}$ (square-root cancellation), applied via the triangle inequality across all $3^l - 1$ nonzero frequencies, requires $j \geq 2 \log_4(3)l \approx 1.585l$.*

Proof. Write $T := |R_{j-1,j}| = \binom{2j}{j}$. Positivity of $N(z)$ for every z via $\sum_{t \neq 0} |S(t)| < T$ combined with the uniform estimate $|S(t)| \lesssim \sqrt{T}$ over the $3^l - 1$ nonzero frequencies forces $3^l \sqrt{T} \lesssim T$, i.e. $T \gtrsim 3^{2l}$, i.e. $j \gtrsim 2 \log_4(3)l$. $\square$

Proposition 5's threshold is *worse* than Corollary 3's already-proven bound: any approach that only delivers generic square-root cancellation across every frequency cannot even match what is already known by other means, let alone improve on it. This rules out the naive version of the Fourier approach cleanly, and directs attention to whether a finer, frequency-dependent treatment (a major/minor-arc split, in the language of the circle method) can do better.

5.2. A computation: the exceptional mass does not stay small. Numerically, $|S(t)|$ is far from uniform: the frequencies of largest magnitude sit close to dyadic rationals of small denominator, $t/3^l \approx p/2^r$ for small r . The natural repair for Proposition 5's ceiling is to excise these major-arc resonances and control the remainder by an averaged bound; whether that remainder stays sparse enough at the scale that matters is a question this section answers computationally, at $l = 18$, with a negative result. A fixed threshold makes the visible spikes look sparse; the threshold $\eta \sim c/\sqrt{T}$ that actually governs the L^1 norm is different and gives a different picture: summing

every one of the 8,014 primitive coefficients exceeding a generous fixed threshold of $0.001 \cdot T$ contributes less than 132 to an L^1 norm of 5226, so over 97% of the mass at this single computed level lives in an exponentially numerous population of near-average-magnitude coefficients. This is evidence against a sparse-exceptional-set repair at accessible l ; it says nothing about every l .

The same clustering near dyadic rationals connects the problem to a 2011 discussion of Terence Tao's, applying Littlewood–Offord and Riesz-product methods to an object built from the same $2^a 3^b$ -type exponential sums, in the closely related setting of a hypothetical non-trivial Collatz cycle [4]; his construction is structurally close to a boundary slice of $R_{j,k}$ with the bottom exponent pinned at 0. His own conclusion, in his own words, is that “once one uses the transcendence theory bound, ... Littlewood–Offord techniques are no longer of much use,” and that he knows “of no way to get a nontrivial separation property between powers of 2 and powers of 3 other than via transcendence theory methods.”

5.3. A direct test of the residues that actually fail. The most direct test examines the residues that actually fail to be covered, one budget step before covering succeeds. At every level checked, their count is tiny (single or low double digits) against an ambient occupancy mean in the hundreds, sitting in cells whose local intensity, at every resolution checked, would assign them probability below e^{-20} under a locally homogeneous model. Holding every $|S(t)|$ fixed while randomizing the phases of conjugate-symmetric pairs collapses the observed extremal statistics to what a generic-phase model predicts; the actual, unscrambled phases do not. At the levels checked, coverage failure is decided by phase structure that magnitude-only methods (uniform bounds, averages, low moments of $S(t)$) do not see.

Proposition 6. $|S(3^{l-1})|/\binom{2m}{m} \rightarrow 1/\sqrt{3}$ as $m \rightarrow \infty$, for the family $R_{m-1,m}$, for every fixed $l \geq 1$.

Proof sketch. $\sum_i 2^{a_i} 3^i \equiv 2^{a_0} \pmod{3}$, so $S(3^{l-1})/\binom{2m}{m}$ depends only on the limiting distribution of the top exponent a_0 's parity as $m \rightarrow \infty$, which converges to the mixture $\frac{1}{3}\omega + \frac{2}{3}\omega^2$ of the two nontrivial cube roots of unity $\omega = e^{2\pi i/3}$ (the limiting gap $2m-1-a_0$ of the maximum of a random m -subset of $\{0, \dots, 2m-1\}$ is geometric with parameter $1/2$, giving $\Pr[a_0 \text{ even}] \rightarrow 1/3$). This mixture equals $-\frac{1}{2} - \frac{i\sqrt{3}}{6}$, of modulus $1/\sqrt{3}$. $\square$

Uniform decay across every nonzero frequency, the premise a naive statement of the Fourier approach might suggest, is false at this explicit family of frequencies.

6. A THEOREM ON THE RESIDUES THAT RESIST COVERAGE

Write $H(l, j)$ for the set of units modulo 3^l not covered by $R_{j-1,j}$, so that $j \geq j^*(l)$ exactly when $H(l, j) = \emptyset$.

Theorem 7 (Holdout doubling inclusion). *For $j \geq l$, $H(l, j+1) \subseteq 2H(l, j) \cap 4H(l, j)$.*

Proof. $R_{j-1,j}$ is the set of values $r = \sum_{i=0}^{j-1} 2^{b_i} 3^i$ over j -subsets $\{b_0 > \dots > b_{j-1}\} \subseteq \{0, \dots, 2j-1\}$, and $R_{j,j+1}$ the same with $(j+1)$ -subsets of $\{0, \dots, 2j+1\}$. Fix $r \in R_{j-1,j}$ with exponent set S , and $s \in \{1, 2\}$. Shifting every exponent up by s gives $S+s \subseteq \{s, \dots, 2j-1+s\} \subseteq \{1, \dots, 2j+1\}$, a j -subset whose induced value is $2^s r$ (a uniform shift of exponents preserves relative rank, so each term's power of 3 is unchanged). Since $S+s$ contains no 0, prepending the new smallest exponent 0 gives a $(j+1)$ -subset of $\{0, \dots, 2j+1\}$: it becomes the new rank- j term, contributing $2^0 \cdot 3^j = 3^j$. So this witness represents $2^s r + 3^j$, and $3^j \equiv 0 \pmod{3^l}$ once $j \geq l$. Hence $2R_{j-1,j} \cup 4R_{j-1,j} \subseteq R_{j,j+1} \pmod{3^l}$ for $j \geq l$, and every $z \notin 2R_{j-1,j} \cap 4R_{j-1,j}$ failing to be represented at budget $j+1$ must already fail at budget j after halving or quartering. $\square$

Corollary 8 (Chain contraction). *If $x, 2x, \dots, 2^{t-1}x \in H(l, j+1)$ (all reduced modulo 3^l , $t \geq 1$), then $x/4, x/2, x, \dots, 2^{t-2}x \in H(l, j)$, a chain of length $t+1$.*

Proof. Apply Theorem 7 to each of the t elements: $2^i x \in H(l, j+1)$ gives $2^{i-1}x, 2^{i-2}x \in H(l, j)$ for $i = 0, \dots, t-1$. The union of these pairs over all i is exactly $\{x/4, x/2, x, \dots, 2^{t-2}x\}$, a doubling chain of length $t+1$. $\square$

Corollary 9 (Run-length bootstrap). *Writing $\text{maxrun}(H)$ for the length of the longest doubling chain contained in H , $j^*(l) \leq j + \text{maxrun}(H(l, j))$ for every $j \geq l$.*

Proof. By Corollary 8, $\text{maxrun}(H(l, j)) \geq \text{maxrun}(H(l, j+1)) + 1$ whenever $H(l, j+1)$ is nonempty, so $\text{maxrun}(H(l, j))$ strictly increases by at least 1 as j decreases from $j^*(l)$; equivalently it strictly decreases by at least 1 at every budget step increasing from l , until it reaches 0, at which point H is empty. $\square$

Empirical Result 10. Corollary 9 is empirically exact, $j^*(l) = j + \text{maxrun}(H(l, j))$, at every (l, j) with $j \geq l+1$ we were able to compute, $l = 5, \dots, 20$ at every budget and l up to 22 at the single budget $j = l+1$, including a case with $|H(l, j)| > 3 \times 10^6$.

The converse direction, that a doubling chain of the observed maximal length is not merely sufficient but necessary for survival to that budget, is not proven, and is false as an unrestricted statement: $1547 \in 2H(7, 8) \cap 4H(7, 8)$, yet $1547 \notin H(7, 9)$ (both of 1547's two budget-9 witnesses span the full exponent range, the one case Theorem 7's argument does not directly control). We return to a restricted, still-open form of this converse in Section 7.

A natural strengthening of Theorem 7, that every individual witness admits a bounded-cost local repair, is also false in its most optimistic form.

Call the *Durfee depth* of a witness $\{a_0 > \dots > a_{j-1}\} \in R_{j-1, j}$ at budget j the count $\#\{i : a_i \geq j\}$, the number of chosen exponents at or above the budget itself.

Empirical Result 11 (Failure of the cost-1 local repair rule). At the budget transition $(l, j) = (6, 10) \rightarrow (7, 11)$, 100 of the 1458 children (residues newly required for coverage at the larger budget) have minimum achievable Durfee depth exceeding their parent's minimum depth by more than one, up to three; at $(7, 11) \rightarrow (8, 12)$, 332 of 4374 children do, likewise up to three. A correspondingly exhaustive check of minimum-edit-distance witness repair finds distances up to 7 at both transitions, against a distance of 1 for the overwhelming majority. This is evidence against a repair cost of exactly 1 at these two transitions; it does not establish that no bounded repair cost exists.

This falsifies, in its originally proposed cost-1 form, a natural conjecture this project had earlier entertained (that coverage propagates via a uniform, cost-1 local repair rule budget by budget); its valid surviving content is Theorem 7 and its corollaries above.

7. FURTHER STRUCTURE OF THE LAST HOLDOUTS

The residues in $H(l, j^*(l) - 1)$, the last to be covered, satisfy several further exact laws. For $W \geq l - 1$, write

$$U(l, W) := \left\{ \sum_{i=0}^{l-1} 2^{\beta_i} 3^i \bmod 3^l \mid W \geq \beta_0 > \dots > \beta_{l-1} \geq 0 \right\},$$

an l -term analogue of $R_{j-1, j}$ built from a fixed term count and a growing exponent ceiling W . For $j \geq l$,

$$(4) \quad R_{j-1, j} \equiv 2^{j-l} U(l, j+l-1) \pmod{3^l},$$

verified computationally with no exception for every (l, j) , $l = 2, \dots, 5$, $j = l, \dots, l+3$: both sides list the same residues. In particular $j^*(l)$ is the smallest $j \geq l$ with $U(l, j+l-1) = (\mathbb{Z}/3^l\mathbb{Z})^*$, and writing $W^*(l) := j^*(l) + l - 2$ for the width one step before that,

$$(5) \quad H(l, j) = 2^{j-l} K(l, j+l-1), \quad K(l, W) := (\mathbb{Z}/3^l\mathbb{Z})^* \setminus U(l, W).$$

Theorem 12 (Last-holdout parity). *Every element of $H(l, j^*(l) - 1)$ is $\equiv 1 \pmod{3}$.*

Proof. $U(l, \cdot)$ is monotone increasing in W , so $K(l, \cdot)$ is monotone decreasing; every $x \in K(l, W^*(l))$ is, by minimality of $j^*(l)$, representable at width $W^*(l) + 1$ but not at $W^*(l)$. Any witness for x at width $W^*(l) + 1$ using only exponents $\leq W^*(l)$ would already witness x at that smaller width, so every such witness has top exponent $\beta_0 = W^*(l) + 1$. Its contribution to the value, at rank 0, is $2^{\beta_0} \cdot 3^0 = 2^{W^*(l)+1}$, so $x \equiv 2^{W^*(l)+1} \equiv (-1)^{W^*(l)+1} \pmod{3}$. By (5), elements of $H(l, j^*(l) - 1)$ are $2^{j^*(l)-1-l}x$ for such x , hence

$$2^{j^*(l)-1-l}x \equiv (-1)^{j^*(l)-1-l} \cdot (-1)^{W^*(l)+1} = (-1)^{2j^*(l)-2} = 1 \pmod{3},$$

using $W^*(l) + 1 = j^*(l) + l - 1$ and that $2j^*(l) - 2$ is always even. $\square$

Empirical Result 13 (Near-extinction bijection). $H(l, j^*(l) - 1) = 2 \cdot \{x \in H(l, j^*(l) - 2) : x \equiv 2 \pmod{3}\}$, exactly, verified at every $l = 2, \dots, 9$ against the exact table with no exception.

The same width mechanism that proves Theorem 12 governs this bijection one step earlier, through the second-highest exponent β_1 of the extremal witness; unlike the top-exponent argument above, pinning β_1 down needs the one-step width recursion $U(l, W + 1) = U(l, W) \cup 2U(l, W) \cup \text{Corner}(l, W + 1)$ (Section 8); we omit the full derivation and record the identity as verified.

Empirical Result 14 ($\times 4$ -only inheritance). Writing $\pi_l : (\mathbb{Z}/3^{l+1}\mathbb{Z})^* \rightarrow (\mathbb{Z}/3^l\mathbb{Z})^*$ for reduction, at every $l = 2, \dots, 8$ checked, every element of $H(l + 1, j^*(l + 1) - 1)$ whose reduction $\pi_l(x)$ lies in $H(l, j^*(l) - 1)$ satisfies $x \equiv 4\pi_l(x) \pmod{3^{l+1}}$; no instance of a $\times 2$ relation occurs. We do not have a proof: Theorem 12 constrains the mod-3 class at each level individually but does not by itself relate a level- $(l+1)$ last holdout to its level- l reduction.

Empirical Result 15 (Mod-9 two-class bound). Writing $J := j^*(l)$, at every $l = 3, \dots, 9$ checked, $H(l, J - 1) \pmod{9} \subseteq \{4^J, 4^{J+1}\} \pmod{9}$, and the class actually occupied is 4^{J+1} , never 4^J , at every level checked. (An earlier stage of this investigation recorded the excluded class backwards; the values above were independently re-verified by three separate implementations before this printing.) The two-class containment follows the same mechanism as Theorem 12 one digit further (the mod-9 class of a threshold witness is $2^{\beta_0} + 3 \cdot 2^{\beta_1} \pmod{9}$, with β_0 pinned and β_1 's parity giving the two options); excluding 4^J specifically is the sharpest concrete open question this part of the investigation produced.

Proposition 16 (Sufficiency of bounded maxrun). *If $\text{maxrun}(H(l, l + 1))$ is bounded by a constant C for all l , then $j^*(l) \leq l + C + 1$.*

Proof. Corollary 9 at $j = l + 1$. $\square$

The converse is not established, and the natural argument for it does not work: reading Corollary 9 downward gives only $\text{maxrun}(H(l, l + 1)) \geq j^*(l) - l - 1$, a *lower* bound on maxrun, which does not force maxrun to stay bounded when $j^*(l) - l$ does. Proving the converse would need an upper bound on maxrun in terms of $j^*(l) - l$, equivalent to a tightness statement for Corollary 9 that Section 8's corner-redundancy question addresses and that is false in the strong, per-witness form: recall 1547 from Section 6.

A naive expectation, that a random set of holdouts at the observed density would have doubling chains far longer than the flat ceiling of 3–4 measured at every level $l = 5, \dots, 22$, does not survive contact with the actual, rapidly declining density of $H(l, l + 1)$. An independence model matched to that actual density tracks the observed sequence within about one unit at every level, including both the one rise and the one fall in it: $\text{maxrun}(H(l, l + 1))$ is typical for the set's density under that model. This does not, by itself, say anything about whether $\text{maxrun}(H(l, l + 1))$ is bounded; Proposition 16 is the only established link between that question and $j^*(l)$'s rate.

The one-step width identity is exact:

$$U(l, W + 1) = U(l, W) \cup 2U(l, W) \cup \text{Corner}(l, W + 1),$$

where $\text{Corner}(l, W + 1)$ consists of the witnesses using both the top exponent $W + 1$ and the bottom exponent 0 simultaneously. At every width $W \geq 2l + 1$ checked exhaustively, $l = 3, \dots, 13$, $\text{Corner}(l, W + 1)$ contributes nothing new: every corner witness's value is already covered without it. If this *corner-redundancy* property holds at every sufficiently large width in general, it would make $j^*(l)$ computable from a single holdout set, a converse to Corollary 9 and a real practical gain for extending Table 1 beyond $l = 23$; it remains unproven, with the only known failures at the boundary width $W = 2l$.

8. DISCUSSION

Two named questions summarize what stands between this paper's results and a sharper theorem. Deciding which of the two classes in Empirical Result 15 is excluded would turn the observed single-class pattern into a proven statement. Proving corner-redundancy at every sufficiently large width would make Corollary 9's bootstrap tight in general, not merely empirically exact at every computed instance, and would settle the converse to Proposition 16 left open above. Neither question is, on its face, as hard as the Weak Covering Conjecture itself; whether either is substantially easier is not known.

A finite exponential sum built from strictly decreasing, exponentially weighted digit tuples is a natural object beyond this specific covering problem: finite-population L -statistics at non-CLT frequencies and weights, Littlewood–Offord theory for structured coefficient sequences, and the analytic side of the Collatz literature via Tao's discussion are all adjacent areas. Three things found here may recur there under different names: generic cancellation capping out below what combinatorial counting already proves, exceptional mass that does not concentrate on a sparse set at the accessible scale, and coverage failure decided by an extremal arithmetic obstruction that a phase-randomization experiment showed magnitude-only methods miss.

The Weak Covering Conjecture itself, and the exact asymptotic rate of $e(l)$, remain open.

9. CODE AND DATA AVAILABILITY

The repository <https://github.com/faculdade/weak-covering-conjecture> is reserved for this paper's code and data, organized by section, and will be populated before submission.

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